AI-Driven Job Transformation by 2030

PROJECT REPORT

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TECHOLAS TECHNOLOGIES

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INTRODUCTION

Artificial Intelligence (AI) is transforming industries worldwide by automating tasks, optimizing operations, and redefining job roles. While AI increases efficiency and productivity, it also introduces automation risks for various professions. Understanding which jobs are highly exposed to automation is essential for workforce planning, career development, and policymaking.

The dataset used in this project analyses the **impact of AI on jobs by 2030**, focusing on automation probability, AI exposure index, salary levels, education levels, and risk categories.

Traditional datasets often fail to present automation risk insights in an interactive and understandable format. Therefore, this project develops an **interactive Excel dashboard** that converts raw job data into meaningful business insights using PivotTables, charts, KPIs, and slicers.

This dashboard helps users quickly identify:

- High-risk job categories
- Average salary distribution
- AI exposure trends
- Education level impact on automation
- Job risk distribution

ABSTRACT

This project presents an **AI Impact on Jobs Dashboard** developed using Microsoft Excel. The dataset contains job roles categorized by automation probability, AI exposure index, education level, salary, and years of experience.

The dataset was cleaned and structured for accurate analysis. Key performance indicators (KPIs) were created to summarize average salary, average automation risk, and the total number of high-risk jobs.

Various visualizations such as line charts, bar charts, stacked charts, and pie charts were used to highlight patterns and trends in automation risk across different job roles and education levels.

The dashboard allows interactive filtering using slicers such as:

- Risk Category
- Education Level
- AI Exposure Index

This project demonstrates how Excel can be used as a powerful Business Intelligence tool for analysing future workforce risks.

OBJECTIVES

- Analyse average salary across job roles
- Calculate average automation risk
- Identify number of jobs in high-risk category
- Study automation risk based on education level
- Analyse AI exposure index distribution
- Understand experience vs automation trends
- Provide interactive filtering options
- Demonstrate Excel dashboard development skills

METHODOLOGY & DATA ANALYSIS

Step 1: Data Importation

The dataset titled **AI_Impact_on_Jobs_2030.csv** was imported into Microsoft Excel. The dataset includes job-level records containing:

- Job Title
- Education Level
- Years of Experience
- Salary
- Automation Probability (2030)
- Risk Category (Low, Medium, High)
- AI Exposure Index

Step 2: Data Cleaning

The following preprocessing steps were performed:

- Checked for missing (NULL/NA) values
- Verified numerical columns (Salary, Automation Probability, AI Exposure Index)
- Standardized categorical fields (Education Level, Risk Category)
- Ensured correct data types for calculations
- Removed duplicate records (if any)

This ensured accuracy in aggregation and KPI calculations.

Step 3: Data Aggregation & KPI Creation

Using PivotTables, the following KPIs were calculated:

- **Average Salary:** 89,358
- **Average Automation Risk:** 0.50

- **Jobs in High-Risk Category:** 740

Aggregation techniques used:

- COUNT of Job Title
- AVERAGE of Salary
- AVERAGE of Automation Probability
- SUM of Automation Probability (2030)
- Category-based grouping (Risk Category & Education Level)

Step 4: Dashboard Construction

The interactive dashboard was created using:

- Pivot Charts (Line, Bar, Pie, Stacked Area)
- KPI Cards (formatted using merged cells & conditional formatting)
- Slicers for filtering:
    - Risk Category
    - Education Level
    - AI Exposure Index
- Structured layout design with consistent theme colours

APPENDIX

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.svm import SVR

data=pd.read_csv(r"C:\Users\ssss\Downloads\archive
(7)\AI_Impact_on_Jobs_2030.csv")
data

from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
data["Job_Title"]=le.fit_transform(data["Job_Title"])
data
X = data.drop("Automation_Probability_2030", axis=1)
y = data["Automation_Probability_2030"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
models = {
    "Linear Regression": LinearRegression(),

    "Decision Tree": DecisionTreeRegressor(
        max_depth=10,
        min_samples_split=10,
        min_samples_leaf=5,
        random_state=42
    ),

    "Random Forest": RandomForestRegressor(
        n_estimators=300,
        max_depth=10,
        min_samples_split=10,
        min_samples_leaf=5,
        random_state=42
    ),

    "Gradient Boosting": GradientBoostingRegressor(
        n_estimators=200,
        learning_rate=0.05,
        max_depth=5,
        random_state=42
    ),

    "SVR": SVR(
        C=10,
```

```
        epsilon=0.1,
        kernel='rbf'
    )
}
results = []

for name, model in models.items():

    model.fit(X_train, y_train)

    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    train_r2 = r2_score(y_train, y_train_pred)
    test_r2 = r2_score(y_test, y_test_pred)
    mae = mean_absolute_error(y_test, y_test_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))

    results.append([
        name,
        round(train_r2, 4),
        round(test_r2, 4),
        round(mae, 4),
        round(rmse, 4)
    ])


results_df = pd.DataFrame(
```

```
    results,
    columns=["Model", "Train R2", "Test R2", "MAE", "RMSE"])
results_df

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix



rf_final = RandomForestRegressor(
    n_estimators=200,
    max_depth=10,
    min_samples_split=5,
    min_samples_leaf=2,
    random_state=42
)
```

	model	**Train R2**	**Test R2**	**MAE**
0	Linear Regression	0.1511	0.1914	0.1883
1	Decision Tree	0.8895	0.8226	0.0855
2	Random Forest	0.9022	0.8636	0.0781
3	Gradient Boosting	0.9124	0.8596	0.0790
4	SVR	0.8852	0.8230	0.0866

RESULTS & VISUALIZATION

KPI Cards

- **Average Salary:** 89,358
- **Average Automation Risk:** 0.50
- **High Risk Jobs:** 740

Charts & Analysis

1. Line Chart – Salary vs Years of Experience

Shows salary trends based on experience levels. Salary generally increases with experience but does not eliminate automation risk.

2. Clustered Bar Chart – Education Level vs Risk Category

Compares High, Medium, and Low risk jobs across education levels. Some education levels show higher concentration in medium risk category.

3. Stacked Area Chart – Risk Category by Experience

Displays how automation risk varies across years of experience.

4. Horizontal Bar Chart – Automation Probability by Job Title

Highlights which job roles are more likely to be automated by 2030.

5. Pie Chart – Job Distribution by Risk Category

Shows overall proportion of High, Medium, and Low risk jobs.

6. Bar Chart – Average AI Exposure Index by Risk Category

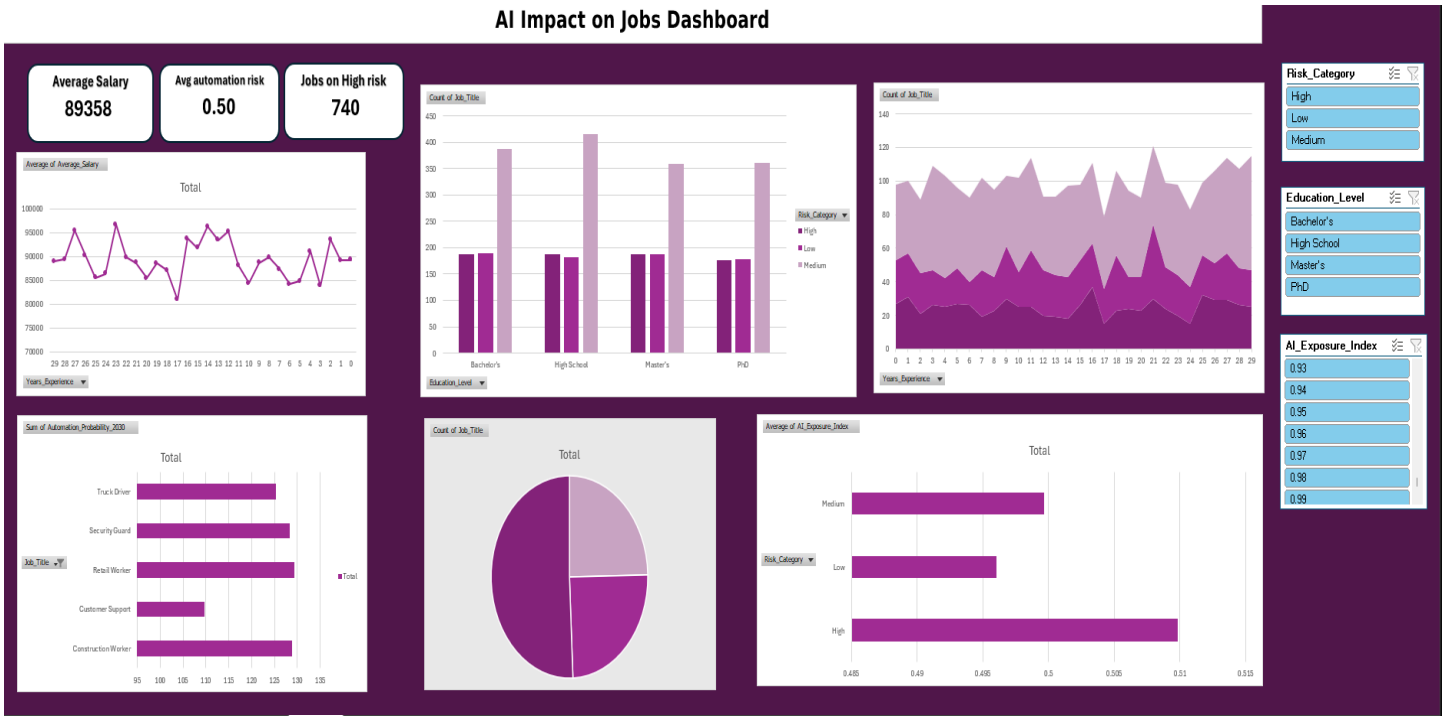
Indicates that High-risk jobs have higher AI exposure index values.

DASHBOARD

The dashboard integrates KPIs, charts, and slicers into a single professional layout.

Interactive filters allow users to:

- Select specific education levels
- Filter by risk category
- Adjust AI exposure index



PROJECT INSIGHTS

1. **Moderate Overall Automation Risk**

   The average automation probability is around **0.50**, indicating that nearly half of the job tasks across the dataset are potentially automatable by 2030.

2. **Large High-Risk Workforce Segment**

   A significant number of jobs (740) fall into the **High-risk category**, suggesting substantial workforce exposure to AI-driven automation.

3. **Strong Link Between AI Exposure and Automation Risk**

   Jobs with higher **AI Exposure Index** values consistently show higher automation probability, confirming that AI intensity directly influences job risk.

4. **Education Reduces but Does Not Eliminate Risk**

   Higher education levels tend to show relatively lower automation risk, but no qualification level is completely immune to AI impact.

5. **Experience Is Not a Guaranteed Protection**

   Even professionals with many years of experience are exposed to automation if their roles involve repetitive or rule-based tasks.

6. **Medium-Risk Jobs Dominate the Dataset**

   Most jobs fall under the **Medium-risk category**, indicating that AI is more likely to transform roles rather than completely replace them.

7. **Nature of Work Matters More Than Salary**

   Automation risk is primarily determined by task type (routine vs analytical/creative), not by salary level.

SUGGESTIONS

Based on the analysis:

- Professionals should upskill in AI-resistant skills such as critical thinking and creativity.
- Organizations should invest in workforce reskilling programs.
- Policy makers should support training initiatives in high-risk sectors.
- Future analysis can integrate industry type for deeper segmentation.
- The dashboard can be migrated to Power BI for real-time updates and predictive analytics.

CONCLUSION

This project successfully analyses the potential impact of Artificial Intelligence on the future workforce using Microsoft Excel as a Business Intelligence tool. By transforming raw job-level data into an interactive dashboard, the study highlights key insights related to automation probability, AI exposure, salary distribution, education level, and risk categorization. The findings indicate that automation risk is not limited to low-skilled roles but affects a wide range of professions depending on the nature of tasks performed. While many jobs fall under the medium-risk category—suggesting role transformation rather than complete replacement—the presence of a significant high-risk segment emphasizes the urgent need for continuous upskilling and workforce adaptation.

Furthermore, the analysis demonstrates that factors such as education and experience reduce risk but do not eliminate exposure to AI-driven changes. The dashboard provides a clear understanding that future job security will depend more on adaptability, technological awareness, and skill enhancement rather than traditional qualifications alone. Overall, this project proves that Excel can effectively be used to perform structured data analysis and create professional dashboards that support informed decision-making in workforce planning, career strategy, and policy development in the era of AI transformation.

REFRENCES: https://www.kaggle.com/datasets/khushikyad001/ai-impact-on-jobs-2030